

\$18.5 billion by 2023, and India is uniquely positioned to capitalise on this growth. With a robust digital infrastructure, a rapidly expanding tech-savvy population, and government-led innovation initiatives, India is on track to become a hub for flexible electronics manufacturing.

However, as highlighted at recent conferences, the future of this industry depends on collaboration. India's emergence as a leader hinges on partnerships between research institutions, industry players, and government entities. One notable success story is Holistic, a company that began by introducing holograms to India and has since evolved into a global provider of anti-counterfeiting solutions. By integrating technologies such as QR codes and RFID, Holistic has demonstrated the vast potential of electronics in ensuring supply chain integrity and combatting counterfeiting.

Sustainable growth, a vision for the future

As the flexible electronics industry expands, sustainability remains a key focus. Innovations in biodegradable electronics and self-healing materials are paving the way for environmentally friendly solutions. Flexible batteries are also being developed to power wearables and other devices, creating new

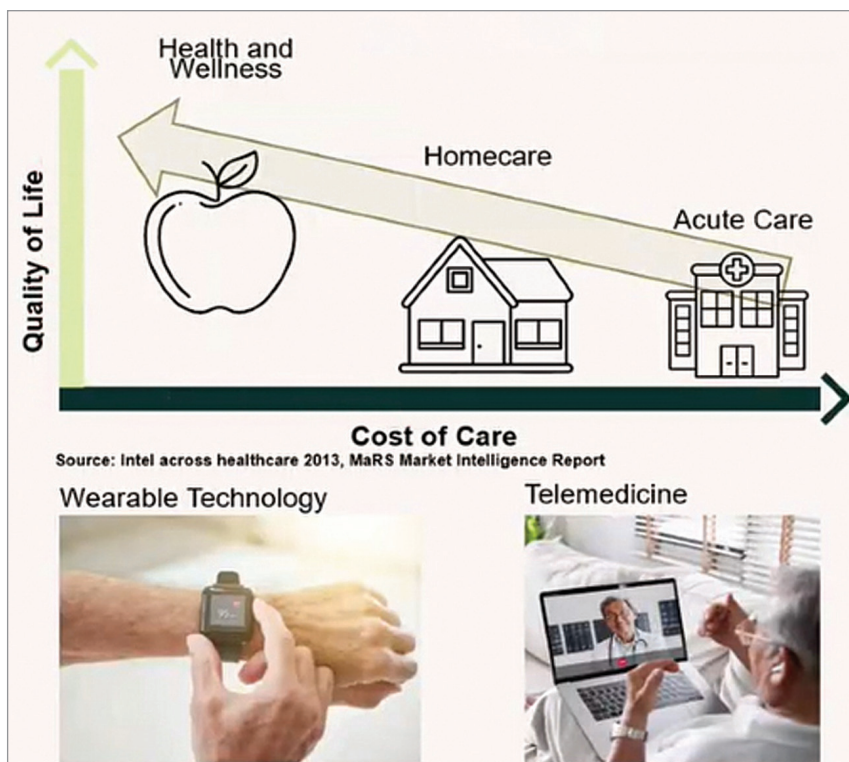


Fig. 4: Decentralisation connectivity

opportunities for energy storage and portable power solutions.

The integration of IoT and AI into this ecosystem will further enhance its capabilities. Wearables that not only track health data but also predict potential issues before they arise are on the horizon. With AI-powered healthcare and smart packaging, India is positioned to revolutionise multiple industries.

The journey towards a future shaped by flexible, printable

electronics is only beginning. India's role in this transformation will depend on collaboration between academia, industry, and government. By aligning research with real-world applications and fostering innovation in manufacturing, India has the potential to become a global leader in the next generation of flexible electronics.

As material scientists, engineers, and manufacturers continue to push technological boundaries, flexible electronics will continue to evolve with limitless potential. With the right investments, infrastructure, and partnerships, the future of flexible electronics is promising—and India is prepared to take the lead. **EFY**

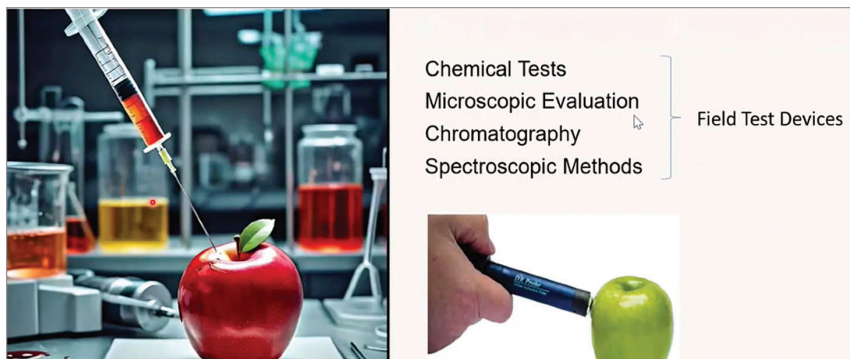


Fig. 3: Growing consciousness about food quality

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